

Project Overview: Recommendations to Improve Network Security

Estimated duration: 5 minutes

Learning objectives

After completing this project, you will be able to evaluate an organization's network infrastructure to identify security weaknesses, propose security enhancements based on infrastructure findings, apply defense-in-depth and least-privilege principles to real-world scenarios, identify common web application vulnerabilities using OWASP guidelines, and recommend practical solutions to mitigate basic OWASP security risks.

Introduction

In today's digital landscape, organizations face an increasing number of sophisticated cyberthreats that target networks, web applications, and sensitive data. As companies expand their digital presence and adopt remote work models, the attack surface grows significantly. Effective network security is no longer optional—it is a critical business requirement. This project focuses on evaluating an organization's existing network infrastructure and proposing practical security improvements to reduce risk, enhance resilience, and protect sensitive information.

Scenario background

Jackson Corporation, a growing organization with both internal and customer-facing systems, has recently experienced a surge in cyberthreat activity. These threats may include unauthorized access attempts, data breaches, malware infections, and web application attacks. Concerned about the potential impact on its operations, reputation, and customer trust, the company has hired a security professional to assess its current infrastructure and recommend improvements.

The organization operates a mixed environment that includes internal networks, web servers, remote users, and custom-developed web applications. While the current setup supports business operations and collaboration, it lacks many essential security controls expected in a modern enterprise environment. The absence of layered defenses, centralized monitoring, and secure development practices places Jackson Corporation at significant risk of both internal and external attacks.

Purpose of the project

The purpose of this project is to analyze Jackson Corporation's existing network and application security posture, identify weaknesses, and propose realistic, effective security enhancements. The project emphasizes a defense-in-depth approach, where multiple security controls work together to protect systems and data.

Specifically, this project aims to:

- Evaluate key components of the organization's network infrastructure
- Identify gaps in security controls and processes
- Recommend improvements that align with industry best practices
- Address common web application vulnerabilities based on OWASP guidelines
- Improve the organization's ability to detect, prevent, and respond to cyber incidents

What is expected in this project

As part of this final project, you are expected to act as a security consultant for Jackson Corporation. Using the findings provided in the scenario, you must analyze each identified issue and document clear, practical security recommendations.

Your responsibilities include:

- Reviewing the current state of the organization's firewall, web servers, network monitoring, breach detection, remote access, software development practices, and web application security
- Explaining why each identified issue presents a security risk to the organization
- Proposing specific and realistic security solutions for each area of concern
- Applying industry best practices and core security principles such as defense in depth and least privilege
- Addressing each OWASP-related issue (A–D) individually with appropriate technical and procedural controls
- Documenting your recommendations clearly and professionally in the provided project document

Your responses should demonstrate an understanding of fundamental cybersecurity concepts and their application in a real-world organizational environment.

Why security improvements are necessary

The findings from the infrastructure review reveal several critical issues:

- A single, poorly segmented firewall increases the risk of attackers moving freely between external and internal networks
- Internally hosted web servers are exposed to unnecessary threats without proper isolation or layered protection
- The lack of centralized network monitoring and breach detection limits visibility into malicious activity and delays incident response
- The absence of secure remote access solutions puts remote employees and corporate resources at risk
- A traditional software development lifecycle that ignores security allows vulnerabilities to reach production systems
- Multiple OWASP Top 10 web application vulnerabilities, such as weak passwords, unencrypted data, and lack of input validation, expose sensitive customer and organizational data

If left unaddressed, these weaknesses could result in data breaches, financial losses, regulatory penalties, and reputational damage. Proactively implementing security controls helps reduce these risks and ensures the organization can operate securely in an increasingly hostile cyber environment.

Scope of the project

This project covers seven major areas of concern:

1. Firewall architecture and traffic segmentation
2. Web server placement and protection
3. Network monitoring and visibility

4. Breach detection and incident awareness
5. Secure remote work access
6. Secure software development practices
7. Web application security improvements based on OWASP guidelines

For each area, practical and achievable recommendations are provided, focusing on improving security while maintaining business efficiency.

Conclusion

This project demonstrates how a structured evaluation of network and application security can uncover critical vulnerabilities and guide effective remediation strategies. By implementing the recommended improvements, Jackson Corporation can significantly strengthen its security posture, protect sensitive data, and better defend against modern cyberthreats. The recommendations provided are designed to be scalable, realistic, and aligned with industry best practices, making them suitable for both immediate and long-term security enhancement.



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